
i.MX RT1060 Processor Reference Manual

With annotations for Teensy
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Chapter 1

About this Document

1.1 Audience

The reference manual is intended for the board-level product designers and product software developers. This manual assumes that the reader has a background in computer engineering and/or software engineering and understands the concepts of the digital system design, microprocessor architecture, input/output (I/O) devices, industry standard communication, and device interface protocols.

1.2 Organization

The reference manual describes the chip at a system level and provides an architectural overview. It also describes the system memory map, system-level interrupt events, external pins and pin multiplexing, external memory, system debug, system boot, multimedia subsystem, power management, and system security.

1.3 Suggested Reading

This section lists additional resources that provide background for the information in the reference manual, as well as general information about the architecture.

Table 1-1. Suggested Reading

Type	Description	Resource
Fact Sheet	The Fact Sheet is an overview of the product key features and its uses.	i.MX RT Series Crossover Processor Fact Sheet
Reference Manual	The Reference Manual contains a comprehensive description of the structure and function (operation) of a device.	This document
Data Sheet	The Data Sheet includes electrical characteristics and signal connections.	i.MX RT1060 Data Sheet - Industrial Products

Table continues on the next page...

Table 1-1. Suggested Reading (continued)

Type	Description	Resource
		i.MX RT1060 Data Sheet - Consumer Products i.MX RT1060 Data Sheet Supplement
Chip Errata	The Chip mask set Errata provides additional or corrective information for a particular device mask set.	i.MX RT1060 Chip Errata
Application Notes	Provides additional information about particular device feature or function.	
	AN12419: Secure JTAG for i.MXRT10xx	AN12419
	AN12085: How to use i.MX RT Low Power Feature	AN12085
	AN12077: Using the i.MX RT FlexRAM	AN12077
	Other Application Notes	Other i.MX RT1060 Application Notes
Web Sites	Product summary page on nxp.com with documentation, software, and resources for the device.	i.MX RT1060 Product Summary Page
	Product documentation page on nxp.com with a list of all documentation related to the device.	i.MX RT1060 Documentation
Community Forum	i.MX RT Community support forum for questions, support, and information about the device.	i.MX RT Community

1.4 Conventions

The reference manual uses the following notational conventions:

cleared / set

When a bit has a value of zero, it is said to be cleared; when it has a value of one, it is said to be set.

mnemonics

Instruction mnemonics are shown in lowercase bold.

italics

Italics indicate variable command parameters, for example, **bcctrx**.

The book titles in the text are set in italics.

15

An integer in decimal.

0x

the prefix to denote a hexadecimal number.

0b

The prefix to denote a binary number. Binary values of 0 and 1 are written without a prefix.

n'H4000CA00

The n -bit hexadecimal number.

BLK_REG_NAME

The register names are all uppercase. The block mnemonic is prepended with an underscore delimiter (_).

BLK_REG[FIELD]

The fields within registers appear in brackets. For example, ESR[RLS] refers to the Receive Last Slot field of the ESAI Status Register.

BLK_REG[n]

The bit number n within the BLK.REG register.

BLK_REG[$l:r$]

The register bit ranges. The ranges are indicated by the left-most bit number l and the right-most bit number r , separated by a colon (:). For example, ESR[15:0] refers to the lower half word in the ESAI Status Register.

x, U

In some contexts, such as signal encodings, an unitalicized x indicates a "don't care" or "uninitialized". The binary value can be 1 or 0.

x

An italicized x indicates an alphanumeric variable.

n, m

Italicized n or m represent integer variables.

!

Binary logic operator NOT.

&&

Binary logic operator AND.

||

Binary logic operator OR.

^ or <O+>

Binary logic operator XOR. For example, $A <O+> B$.

|

Bit-wise OR. For example, $0b0001 | 0b1000$ yields the value of $0b1001$.

&

Bit-wise AND. For example, $0b0001 \& 0b1000$ yields the value of $0b0000$.

{A,B}

Concatenation, where the n -bit value A is prepended to the m -bit value B to form an $(n+m)$ -bit value. For example, $\{0, \text{REG}_m[14:0]\}$ yeilds a 16-bit value with 0 in the most significant bit.

- or grey fill

Indicates a reserved bit field in a register. Although these bits can be written to with ones or zeros, they always read zeros.

>>

Shift right logical one position.

<<
Shift left logical one position.

<=
Assignment.

==
Compare equal.

!=
Compare not equal.

>
Greater than.

<
Less than.

1.5 Register Access

1.5.1 Register Diagram Field Access Type Legend

This figure provides the interpretation of the notation used in the register diagrams for a number of common field access types:

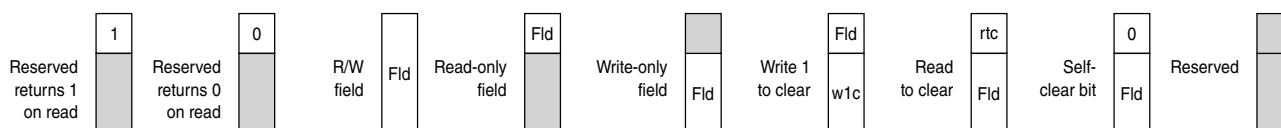


Figure 1-1. Register Field Conventions

NOTE

For reserved register fields, the software should mask off the data in the field after a read (the software can't rely on the contents of data read from a reserved field) and always write all zeros.

1.5.2 Register Macro Usage

A common operation is to update one field without disturbing the contents of the remaining fields in the register. Normally, this requires a read-modify-write (RMW) operation, where the CPU reads the register, modifies the target field, then writes the results back to the register. This is an expensive operation in terms of CPU cycles, because of the initial register read.

To address this issue, some hardware registers are implemented as a group, including registers that can be used to either set, clear, or toggle (SCT) individual bits of the primary register. When writing to an SCT register, all the bits set to 1 perform the associated operation on the primary register, while the bits set to 0 are not affected. The SCT registers always read back 0, and should be considered write-only. The SCT registers are not implemented if the primary register is read-only.

With this architecture, it is possible to update one or more fields using only register writes. First, all bits of the target fields are cleared by a write to the associated clear register, then the desired value of the target fields is written to the set register. This sequence of two writes is referred to as a clear-set (CS) operation.

A CS operation does have one potential drawback. Whenever a field is modified, the hardware sees a value of 0 before the final value is written. For most fields, passing through the 0 state is not a problem. Nonetheless, this behavior is something to consider when using a CS operation.

Also, a CS operation is not required for fields that are one-bit wide. While the CS operation works in this case, it is more efficient to simply set or clear the target bit (that is, one write instead of two). A simple set or clear operation is also atomic, while a CS operation is not.

Note that not all macros for set, clear, or toggle (SCT) are atomic. For registers that do not provide hardware support for this functionality, these macros are implemented as a sequence of read-modify-write operations. When an atomic operation is required, the developer should pay attention to this detail, because unexpected behavior might result if an interrupt occurs in the middle of the critical section comprising the update sequence.

A set of SCT registers is offered for registers in many modules on this device, as described in this manual. In a module memory map table, the suffix `_SET`, `_CLR`, or `_TOG` is added to the base name of the register. For example, the `CCM_ANALOG_PLL_ARM` register has three other registers called `CCM_ANALOG_PLL_ARM_SET`, `CCM_ANALOG_PLL_ARM_CLR`, and `CCM_ANALOG_PLL_ARM_TOG`.

In the sub-section that describes one of these sets of registers, a short-hand convention is used to denote that a register has the SCT register set. There is an italicized *n* appended to the end of the short register name. Using the above example, the name used for this register is `CCM_ANALOG_PLL_ARMn`. When you see this designation, there is a SCT register set associated with the register, and you can verify this by checking it in the memory map table. The address offset for each of these registers is given in the form of the following example:

Address: `20C_8000h` base + `0h` offset + $(4d \times i)$, where $i=0d$ to $3d$

Acronyms and Abbreviations

In this example, the address for each of the base registers and their three SCT registers can be calculated as:

Register	Address
CCM_ANALOG_PLL_ARM	20C_8000h
CCM_ANALOG_PLL_ARM_SET	20C_8004h
CCM_ANALOG_PLL_ARM_CLR	20C_8008h
CCM_ANALOG_PLL_ARM_TOG	20C_800Ch

1.6 Acronyms and Abbreviations

The table below contains acronyms and abbreviations used in this document.

Acronyms and Abbreviated Terms

Term	Meaning
ACMP	Analog Comparator
ADC	Analog-to-Digital Converter
AHB	Advanced High-performance Bus
AIPS	Arm IP Bus
ALU	Arithmetic Logic Unit
AMBA	Advanced Microcontroller Bus Architecture
APB	Advanced Peripheral Bus
ASRC	Asynchronous Sample Rate Converter
AXI	Advanced eXtensible Interface
BEE	Bus Encryption Engine
CA/CM	Arm Cortex-A/Cortex-M
CAN	Controller Area Network
CCM	Clock Controller Module
CM7	Arm Cortex M7 Core
CPU	Central Processing Unit
CSI	CMOS Sensor Interface
CSU	Central Security Unit
CTI	Cross Trigger Interface
DAP	Debug Access Port
DCP	Data Co-Processor
DDR	Double data rate
DMA	Direct memory access
DPLL	Digital phase-locked loop
DRAM	Dynamic random access memory

Table continues on the next page...

Term	Meaning
ECC	Error correcting codes
LPSPi	Low-power SPI
EDMA	Enhanced Direct Memory Access
EIM	External Interface Module
ENET	Ethernet
EPIT	Enhanced Periodic Interrupt Timer
EPROM	Erasable Programmable Read-Only Memory
ETF	Embedded Trace FIFO
ETM	Embedded Trace Macrocell
FIFO	First-In-First-Out
GIC	General Interrupt Controller
GPC	General Power Controller
GPIO	General-Purpose I/O
GPR	General-Purpose Register
GPS	Global Positioning System
GPT	General-Purpose Timer
GPU	Graphics Processing Unit
GPV	Global Programmers View
HAB	High-Assurance Boot
I2C or I ² C	Inter-Integrated Circuit
IC	Integrated Circuit
IEEE	Institute of Electrical and Electronics Engineers
IOMUX	Input-Output Multiplexer
IP	Intellectual Property
IrDA	Infrared Data Association
JTAG	Joint Test Action Group (a serial bus protocol usually used for test purposes)
ELCDIF	Liquid Crystal Display Interface
LDO	Low-Dropout
LIFO	Last-In-First-Out
LRU	Least-Recently Used
LSB	Least-Significant Byte
LUT	Look-Up Table
LVDS	Low Voltage Differential Signaling
MAC	Medium Access Control
MCM	Miscellaneous Control Module
MMC	Multimedia Card
MSB	Most-Significant Byte
MT/s	Mega Transfers per second
OCRAM	On-Chip Random-Access Memory
OCOTP	On-Chip One-Time Programmable Controller
PCI	Peripheral Component Interconnect

Table continues on the next page...

Acronyms and Abbreviations

Term	Meaning
PCIe	PCI express
PGC	Power Gating Controller
PIC	Programmable Interrupt Controller
PMU	Power Management Unit
POR	Power-On Reset
PSRAM	Pseudo-Static Random Access Memory
PWM	Pulse Width Modulation
PXP	Pixel Pipeline
QoS	Quality of Service
R2D	Radians to Degrees
RISC	Reduced Instruction Set Computing
ROM	Read-Only Memory
RTOS	Real-Time Operating System
Rx	Receive
SAI	Synchronous Audio Interface
SD	Secure Digital
SDIO	Secure Digital Input/Output
SDLC	Synchronous Data Link Control
SDMA	Smart DMA
SIM	Subscriber Identification Module
SNVS	Secure Non-Volatile Storage
SoC	System-on-Chip
SPBA	Shared Peripheral Bus Arbiter
SPDIF	Sony Philips Digital Interface
SPI	Serial Peripheral Interface
SRAM	Static Random-Access Memory
SRC	System Reset Controller
TFT	Thin-Film Transistor
TPIU	Trace Port Interface Unit
TSGEN	Time Stamp Generator
Tx	Transmit
TZASC	TrustZone Address Space Controller
UART	Universal Asynchronous Receiver/Transmitter
USB	Universal Serial Bus
USDHC	Ultra Secured Digital Host Controller
WDOG	Watchdog
WLAN	Wireless Local Area Network
WXGA	Wide Extended Graphics Array

Chapter 2

Introduction

2.1 Introduction

The i.MX RT1060 is a new processor family featuring NXP's advanced implementation of the high performance Arm Cortex[®]-M7 Core. It offers high-performance processing optimized for lowest power consumption and best real-time response.

This device provides various memory interfaces, including SDRAM, Raw NAND FLASH, NOR FLASH, SD/eMMC, Quad SPI (FlexSPI), and a wide range of other interfaces for connecting peripherals, such as WLAN, Bluetooth[™], displays, camera sensors, and GPS. Same as other i.MX processors, this i.MX RT series also has rich audio and video features, including LCD display, basic 2D graphics, camera interface, SPDIF and I2S audio interface.

2.1.1 Block Diagram

The functional block diagram is shown in the figure below. This diagram provides a view of the chip's major functional components and core complexes.

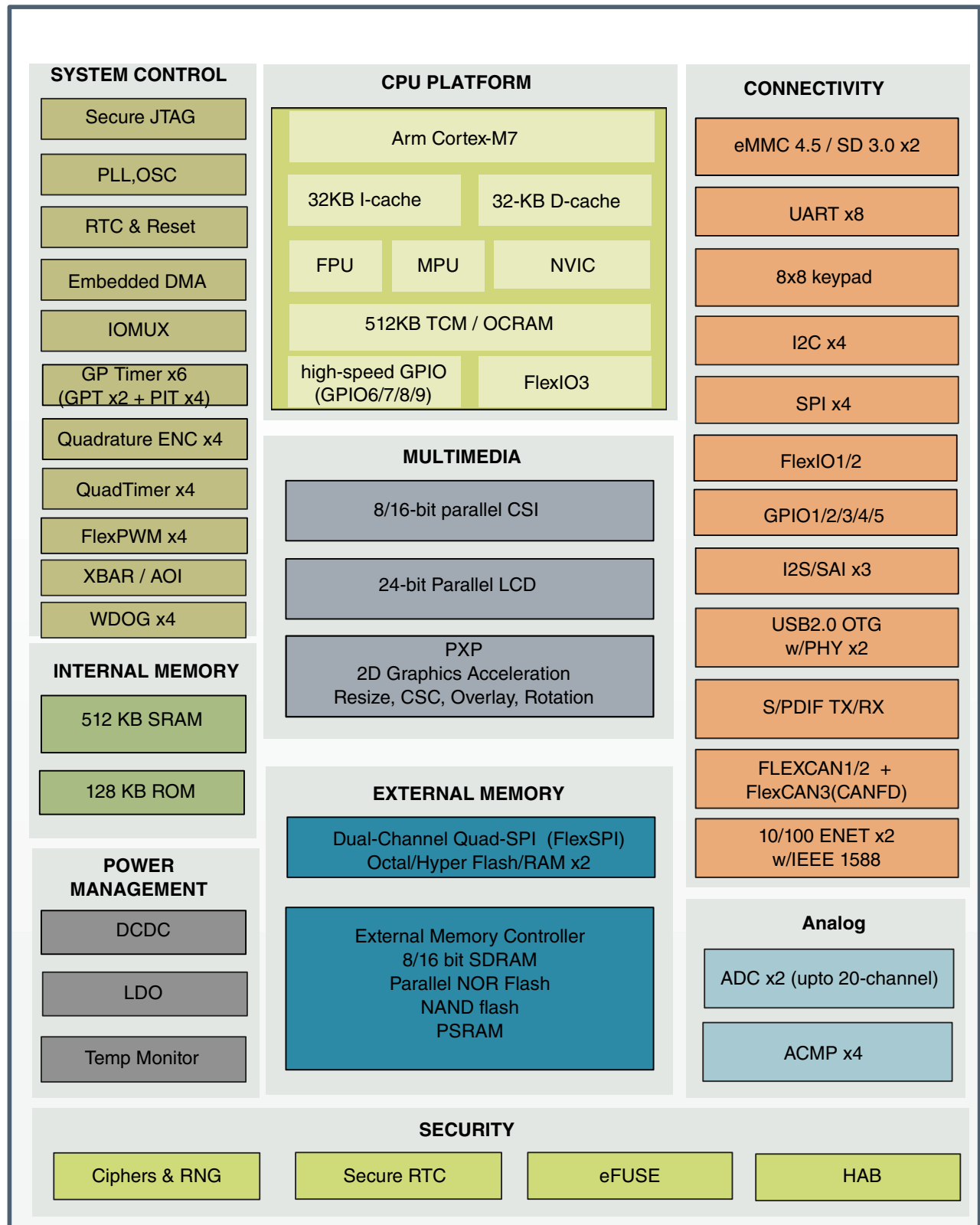


Figure 2-1. Simplified Block Diagram

2.2 Features

The i.MX RT1060 processors are based on Arm®Cortex®-M7 Platform, which have the following features:

Single Arm Cortex-M7 with:

- 32KB L1 Instruction Cache
- 32KB L1 Data Cache
- Single-precision and double-precision FPU (Floating Point Unit)
- Integrated Memory Protection Unit (MPU), up to 16 individual protection regions
- Tightly coupled GPIOs (GPIO6/7/8/9), operating at the same frequency as Arm
- Up to 512KB I-TCM and D-TCM in total

On Chip Memory:

- Boot ROM (128KB)
- On-chip RAM, configurable RAM up to 1MB (512KB OCRAM shared between ITCM/DTCM and OCRAM, as well as dedicated 512KB OCRAM)

External Memory Interfaces:

- 8/16-bit SDRAM, up to SDRAM-166
- 8-bit SLC NAND FLASH, with ECC handled in software
- SD/eMMC
- ×2 Single/Dual channel Quad SPI FLASH with XIP support
- Parallel NOR FLASH with XIP support
- SPI NOR/NAND FLASH

Graphics:

- Generic 2D Graphics engine (PXP)
 - BitBlit
 - Flexible image composition options – alpha, chroma key
 - Image rotation (90, 180, 270 degree rotation)
 - Porter-Duff operation
 - Image resize
 - Color space conversion
 - Multiple pixel format support (RGB, YUV444, YUV422, YUV420, YUV400)
 - Standard 2D-DMA operation

Display Interface:

- Parallel RGB LCD interface
 - Support 8-/16-/24-bit interface
 - Support up to WXGA resolution

Features

- Smart LCD Display with 8/16-bit MPU/8080 interface
- Support Index color with 256 entry × 24 bit color LUT

Camera Sensor Interface:

- Support 8/16/24-bit CSI Input

Audio:

- SPDIF Input and Output
- 3x SAI (synchronous audio interface) modules supporting I2S, AC97, TDM, and Codec/DSP interfaces
- MQS interface for medium quality audio via GPIO pads

Connectivity:

- 2x USB 2.0 OTG controller with integrated PHY interfaces
- 2x Ultra Secure Digital Host Controller (uSDHC) interfaces
- 2× 10M/100M Ethernet controller with support for IEEE1588
- 8x Universal asynchronous receiver/transmitter (UARTs) modules
- 4x I2C modules
- 4x SPI modules
- 2x FlexCAN modules
- 1x FlexCAN module with Flexible Data-rate support (CANFD/FlexCAN3)
- 3x FlexIO modules

Timers:

- 2x General Programmable Timer (GPT)
- 4-channel Periodical Interrupt Timer (PIT)
- 4x Quad Timer (QTimer)
- 4x FlexPWM

Analog:

- 2x Analog-Digital-Converters (ADC) (up to 20 channels)
- 4x Analog Comparators (ACMP)

Security:

- High Assurance Boot (HAB)
- Data Co-Processor (DCP)
 - AES-128, ECB, and CBC mode
 - SHA-1 and SHA-256
 - CRC-32
- Bus Encryption Engine (BEE)
 - AES-128, ECB, and CTR mode
 - On-the-fly QSPI Flash decryption
- True Random Number Generator (TRNG)
- Secure Non-volatile Storage (SNVS)

- Secure real-time clock (RTC)
- Secure JTAG Controller (SJC)

System Debug:

- Arm CoreSight debug and trace architecture
- Trace Port Interface Unit (TPIU) to support off-chip real-time trace
- Cross Triggering Interface (CTI)
- Support for 5-pin (JTAG) and SWD debug interfaces

Power Management:

- Full PMIC integration, including on-chip DCDC and LDO
- Temperature sensor with programmable trim points
- GPC hardware power management controller

2.3 Target Applications

This processor can be used in areas such as industrial HMI, IoT, motor control and home appliances, etc.

The architecture's flexibility enables it to be used in a wide variety of other general embedded applications too. This processor provides all interfaces necessary to connect peripherals such as WLAN, Bluetooth[™], GPS, camera sensors, and multiple displays.

2.4 Endianness Support

The chip supports only the Little Endian mode.

Chapter 3

Memory Maps

3.1 Memory system overview

This section introduces the memory architecture of the chip.

NOTE

Accessing the reserved memory regions can result in unpredictable behavior.

3.2 Arm Platform Memory Map

The chip memory map has been provided in the following tables.

Table 3-1. System memory map (CM7)

Start Address	End Address	Size	Description
E010_0000	FFFF_FFFF	511MB	Reserved
E000_0000	E00F_FFFF	1MB	CM7 PPB ARM peripherals
8000_0000	DFFF_FFFF	1.5GB	SEMC external memories (SDRAM, NOR, PSRAM, NAND and 8080) shared memory space
7FC0_0000	7FFF_FFFF	4MB	FlexSPI RX FIFO
7F80_0000	7FBF_FFFF	4MB	FlexSPI TX FIFO
7F40_0000	7F7F_FFFF	4MB	FlexSPI2 RX FIFO
7F00_0000	7F3F_FFFF	4MB	FlexSPI2 TX FIFO
7000_0000	7EFF_FFFF	240MB	FlexSPI2/ FlexSPI2 ciphertext PSRAM (EXTMEM)
6000_0000	6FFF_FFFF	256MB	FlexSPI/ FlexSPI ciphertext Flash (PROGMEM, FLASHMEM)
4800_0000	5FFF_FFFF	384MB	Reserved
4400_0000	47FF_FFFF	64MB	Reserved
4200_00004210_0000	43FF_FFFF	32MB31MB	Reserved
4200_0000	420F_FFFF	1MB	AIPS-5 Low latency peripherals (sadly, no DMA access)

Table continues on the next page...

Table 3-1. System memory map (CM7) (continued)

Start Address	End Address	Size	Description
4180_0000	41FF_FFFF	8MB	Reserved
4170_0000	417F_FFFF	1MB	GPV Reserved
4160_0000	416F_FFFF	1MB	GPV Reserved
4150_0000	415F_FFFF	1MB	GPV Reserved
4140_0000	414F_FFFF	1MB	"cpu" configuration port
4130_0000	413F_FFFF	1MB	Reserved for "ems" GPV
4120_0000	412F_FFFF	1MB	Reserved for "per" GPV
4110_0000	411F_FFFF	1MB	"m" configuration port
4100_0000	410F_FFFF	1MB	"main" configuration port
4040_0000	40FF_FFFF	12MB	Reserved
4030_0000	403F_FFFF	1MB	AIPS-4
4020_0000	402F_FFFF	1MB	AIPS-3
4010_0000	401F_FFFF	1MB	AIPS-2
4000_0000	400F_FFFF	1MB	AIPS-1
3000_0000	3FFF_FFFF	256MB	Reserved
2040_0000	2FFF_FFFF	252MB	Reserved
2030_0000	203F_FFFF	1MB	OCRAM Reserved
2028_0000	202F_FFFF	512KB	OCRAM - FlexRAM
2020_0000	2027_FFFF	512KB	OCRAM2
2010_0000	201F_FFFF	1MB	Reserved
2008_0000	200F_FFFF	512KB	DTCM Reserved
2000_0000	2007_FFFF	512KB	DTCM
1000_0000	1FFF_FFFF	256MB	SEMC (Aliased)Reserved
0800_0000	0FFF_FFFF	128MB	FlexSPI (Aliased)Reserved
0040_0000	07FF_FFFF	124MB	Reserved
0028_0000	003F_FFFF	1536KB	Reserved
0021_80000022_0000	0027_FFFF	416KB384KB	Reserved
0020_0000	0021_7FFF0021_FFFF	96KB128KB	Reserved
0010_0000	001F_FFFF	1MB	ITCM Reserved
0008_0000	000F_FFFF	512KB	ITCM Reserved
0000_0000	0007_FFFF	512KB	ITCM

Teensy startup code automatically partitions RAM1 between ITCM and DTCM depending on compiled code size, by using IOMUXC_GPR_GPR17 (page 363).

The table below shows the Arm IP Bus (AIPS) detailed memory map.

Table 3-2. AIPS-1 memory map

Start Address	End Address	Region	Size	NIC Port
400F_C000	400F_FFFF	AIPS-1	16KB	CCM(CCM)

Table continues on the next page...

Table 3-2. AIPS-1 memory map (continued)

Start Address	End Address	Region	Size	NIC Port
400F_8000	400F_BFFF		16KB	SRC(SRC)
400F_4000	400F_7FFF		16KB	GPC
400F_0000	400F_3FFF		16KB	Reserved
400E_C000	400E_FFFF		16KB	DMA_CH_MUX
400E_8000	400E_BFFF		16KB	EDMA
400E_4000	400E_7FFF		16KB	SJC
400E_0000	400E_3FFF		16KB	TSC_DIG
400D_C000	400D_FFFF		16KB	CSU
400D_8000	400D_BFFF		16KB	ANALOG
400D_4000	400D_7FFF		16KB	SNVS_HP
400D_0000	400D_3FFF		16KB	WDOG2
400C_C000	400C_FFFF		16KB	TRNG
400C_8000	400C_BFFF		16KB	ADC2
400C_4000	400C_7FFF		16KB	ADC1
400C_0000	400C_3FFF		16KB	GPIO5
400B_C000	400B_FFFF		16KB	WDOG3
400B_8000	400B_BFFF		16KB	WDOG1
400B_4000	400B_7FFF		16KB	EWM
400B_0000	400B_3FFF		16KB	CM7_MXRT (FLEXRAM)
400A_C000	400A_FFFF		16KB	IOMUXC_GPR
400A_8000	400A_BFFF		16KB	IOMUXC_SNVS
400A_4000	400A_7FFF		16KB	IOMUXC_SNVS_GPR
400A_0000	400A_3FFF		16KB	Reserved
4009_C000	4009_FFFF		16KB	Reserved
4009_8000	4009_BFFF		16KB	Reserved
4009_4000	4009_7FFF		16KB	ACMP
4009_0000	4009_3FFF		16KB	Reserved
4008_C000	4008_FFFF		16KB	Reserved
4008_8000	4008_BFFF		16KB	Reserved
4008_4000	4008_7FFF		16KB	PIT
4008_0000	4008_3FFF		16KB	DCDC
4007_C000	4007_FFFF		16KB	AIPS-1 Configuration
4004_0000	4007_BFFF		240KB	Reserved
4000_0000	4003_FFFF		256KB	Reserved

The table below shows the AIPS-2 detailed memory map.

Table 3-3. AIPS-2 memory map

Start Address	End Address	Region	Size	NIC Port
401F_C000	401F_FFFF	AIPS-2	16KB	KPP
401F_8000	401F_BFFF		16KB	IOMUXC
401F_4000	401F_7FFF		16KB	OCOTP
401F_0000	401F_3FFF		16KB	GPT2
401E_C000	401E_FFFF		16KB	GPT1
401E_8000	401E_BFFF		16KB	QTimer4
401E_4000	401E_7FFF		16KB	QTimer3
401E_0000	401E_3FFF		16KB	QTimer2
401D_C000	401D_FFFF		16KB	QTimer1
401D_8000	401D_BFFF		16KB	CANFD (FlexCAN3)
401D_4000	401D_7FFF		16KB	FlexCAN2
401D_0000	401D_3FFF		16KB	FlexCAN1
401C_C000	401C_FFFF		16KB	Reserved
401C_8000	401C_BFFF		16KB	Reserved
401C_4000	401C_7FFF		16KB	GPIO4
401C_0000	401C_3FFF		16KB	GPIO3
401B_C000	401B_FFFF		16KB	GPIO2
401B_8000	401B_BFFF		16KB	GPIO1
401B_4000	401B_7FFF		16KB	Reserved
401B_0000	401B_3FFF		16KB	FlexIO2
401A_C000	401A_FFFF		16KB	FlexIO1
401A_8000	401A_BFFF		16KB	Reserved
401A_4000	401A_7FFF		16KB	Reserved
401A_0000	401A_3FFF		16KB	LPUART8
4019_C000	4019_FFFF		16KB	LPUART7
4019_8000	4019_BFFF		16KB	LPUART6
4019_4000	4019_7FFF		16KB	LPUART5
4019_0000	4019_3FFF		16KB	LPUART4
4018_C000	4018_FFFF		16KB	LPUART3
4018_8000	4018_BFFF		16KB	LPUART2
4018_4000	4018_7FFF		16KB	LPUART1
4018_0000	4018_3FFF		16KB	Reserved
4017_C000	4017_FFFF		16KB	AIPS-2 Configuration
4014_0000	4017_BFFF		240KB	Reserved
4010_0000	4013_FFFF		256KB	Reserved

The table below shows the AIPS-3 detailed memory map.

Table 3-4. AIPS-3 memory map

Start Address	End Address	Region	Size	NIC Port
402F_C000	402F_FFFF	AIPS-3	16KB	DCP
402F_8000	402F_BFFF		16KB	Reserved
402F_4000	402F_7FFF		16KB	Reserved
402F_0000	402F_3FFF		16KB	SEMC
402E_C000	402E_FFFF		16KB	Reserved
402E_8000	402E_BFFF		16KB	Reserved
402E_4000	402E_7FFF		16KB	Reserved
402E_0000	402E_3FFF		16KB	USB(USB)
402D_C000	402D_FFFF		16KB	Reserved
402D_8000	402D_BFFF		16KB	ENET
402D_4000	402D_7FFF		16KB	ENET2
402D_0000	402D_3FFF		16KB	Reserved
402C_C000	402C_FFFF		16KB	Reserved
402C_8000	402C_BFFF		16KB	Reserved
402C_4000	402C_7FFF		16KB	USDHC2
402C_0000	402C_3FFF		16KB	USDHC1
402B_C000	402B_FFFF		16KB	CSI
402B_8000	402B_BFFF		16KB	LCDIF
402B_4000	402B_7FFF		16KB	PXP
402B_0000	402B_3FFF		16KB	Reserved
402A_C000	402A_FFFF		16KB	Reserved
402A_8000	402A_BFFF		16KB	FlexSPI
402A_4000	402A_7FFF		16KB	ReservedFlexSPI2
402A_0000	402A_3FFF		16KB	Reserved
4029_C000	4029_FFFF		16KB	Reserved
4029_8000	4029_BFFF		16KB	Reserved
4029_4000	4029_7FFF		16KB	Reserved
4029_0000	4029_3FFF		16KB	Reserved
4028_C000	4028_FFFF		16KB	Reserved
4028_8000	4028_BFFF		16KB	Reserved
4028_4000	4028_7FFF		16KB	Reserved
4028_0000	4028_3FFF		16KB	Reserved
4027_C000	4027_FFFF		16KB	AIPS-3 Configuration
4024_0000	4027_BFFF		240KB	Reserved
4020_0000	4023_FFFF		256KB	Reserved

The table below shows the AIPS-4 detailed memory map.

Table 3-5. AIPS-4 memory map

Start Address	End Address	Region	Size	NIC Port
403F_C000	403F_FFFF	AIPS-4	16KB	LPI2C4
403F_8000	403F_BFFF		16KB	LPI2C3
403F_4000	403F_7FFF		16KB	LPI2C2
403F_0000	403F_3FFF		16KB	LPI2C1
403E_C000	403E_FFFF		16KB	BEE
403E_8000	403E_BFFF		16KB	FLEXPWM4
403E_4000	403E_7FFF		16KB	FLEXPWM3
403E_0000	403E_3FFF		16KB	FLEXPWM2
403D_C000	403D_FFFF		16KB	FLEXPWM1
403D_8000	403D_BFFF		16KB	Reserved
403D_4000	403D_7FFF		16KB	QDC4
403D_0000	403D_3FFF		16KB	QDC3
403C_C000	403C_FFFF		16KB	QDC2
403C_8000	403C_BFFF		16KB	QDC1
403C_4000	403C_7FFF		16KB	XBAR3
403C_0000	403C_3FFF		16KB	XBAR2
403B_C000	403B_FFFF		16KB	XBAR1
403B_8000	403B_BFFF		16KB	AOI2
403B_4000	403B_7FFF		16KB	AOI1
403B_0000	403B_3FFF		16KB	ADC_ETC
403A_C000	403A_FFFF		16KB	Reserved
403A_8000	403A_BFFF		16KB	Reserved
403A_4000	403A_7FFF		16KB	Reserved
403A_0000	403A_3FFF		16KB	LPSPi4
4039_C000	4039_FFFF		16KB	LPSPi3
4039_8000	4039_BFFF		16KB	LPSPi2
4039_4000	4039_7FFF		16KB	LPSPi1
4039_0000	4039_3FFF		16KB	Reserved
4038_C000	4038_FFFF		16KB	SAI3
4038_8000	4038_BFFF		16KB	SAI2
4038_4000	4038_7FFF		16KB	SAI1
4038_0000	4038_3FFF		16KB	SPDIF
4037_C000	4037_FFFF		16KB	AIPS-4 Configuration
4034_0000	4037_BFFF		240KB	Reserved
4030_0000	4033_FFFF		256KB	Reserved

The table below shows the AIPS-5 detailed memory map.

Table 3-6. AIPS-5 memory map No DMA access

Start Address	End Address	Region	Size	NIC Port
4208_0000	420F_FFFF	AIPS-5	512KB	Reserved off platform slots
4207_C000	4207_FFFF		16KB	Reserved
4207_8000	4207_BFFF		16KB	Reserved
4207_4000	4207_7FFF		16KB	Reserved
4207_0000	4207_3FFF		16KB	Reserved
4206_C000	4206_FFFF		16KB	Reserved
4206_8000	4206_BFFF		16KB	Reserved
4206_4000	4206_7FFF		16KB	Reserved
4206_0000	4206_3FFF		16KB	Reserved
4205_C000	4205_FFFF		16KB	Reserved
4205_8000	4205_BFFF		16KB	Reserved
4205_4000	4205_7FFF		16KB	Reserved
4205_0000	4205_3FFF		16KB	Reserved
4204_C000	4204_FFFF		16KB	Reserved
4204_8000	4204_BFFF		16KB	Reserved
4204_4000	4204_7FFF		16KB	Reserved
4204_0000	4204_3FFF		16KB	Reserved
4203_C000	4203_FFFF		16KB	Reserved
4203_8000	4203_BFFF		16KB	Reserved
4203_4000	4203_7FFF		16KB	Reserved
4203_0000	4203_3FFF		16KB	Reserved
4202_C000	4202_FFFF		16KB	Reserved
4202_8000	4202_BFFF		16KB	Reserved
4202_4000	4202_7FFF		16KB	Reserved
4202_0000	4202_3FFF		16KB	FlexIO3
4201_C000	4201_FFFF		16KB	Reserved
4201_8000	4201_BFFF		16KB	Reserved
4201_4000	4201_7FFF		16KB	Reserved
4201_0000	4201_3FFF		16KB	Reserved
4200_C000	4200_FFFF		16KB	GPIO9
4200_8000	4200_BFFF		16KB	GPIO8
4200_4000	4200_7FFF		16KB	GPIO7
4200_0000	4200_3FFF		16KB	GPIO6

The table below shows the PPB detailed memory map.

Table 3-7. PPB memory map

Start Address	End Address	Size	Allocation
E00F_F000	E00F_FFFF	4KB	PPB ROM
E00F_E000	E00F_EFFF	4KB	Processor ROM
E00F_D000	E00F_DFFF	4KB	SYS ROM
E00F_0000	E00F_CFFF	52KB	PPB Reserved
E008_1000	E00E_FFFF	444KB	PPB Reserved
E008_0000	E008_0FFF	4KB	MCM
E004_5000	E007_FFFF	236KB	PPB Reserved
E004_4000	E004_4FFF	4KB	PPB RES
E004_3000	E004_3FFF	4KB	TSGEN
E004_2000	E004_2FFF	4KB	CTI
E004_1000	E004_1FFF	4KB	ETM
E004_0000	E004_0FFF	4KB	TPIU

NOTE

Accessing the reserved memory regions can result in unpredictable behavior.

Chapter 4

Interrupts, DMA Events, and XBAR Assignments

4.1 Chip-specific Interrupt information

Table 4-1. Reference links to related information

Topic	Related module or subsystem	Reference
System memory map	-	System Memory Map
Clocking	CCM	Clock Management Clock Control Module (CCM)
Audio Subsystem	Audio Subsystem	Audio Subsystem Overview
Power management	PMU	Power Management Power Management Unit
Signal multiplexing	IOMUX	External Signals and Pin Multiplexing IOMUX
Interrupts, DMA Events and XBAR Assignments	-	Interrupts, DMA Events and XBAR Assignments

4.2 Overview

This section describes the Interrupt assignments, DMA events, and XBAR resource assignments.

4.3 CM7 interrupts

The Nested Vectored Interrupt Controller (NVIC) collects interrupt request sources and provides an interface to the Cortex-M7 core.

The table below describes the Cortex-M7 interrupt sources:

NOTE

Due to the clock frequency difference between CPU and peripheral, in some corner case, peripheral interrupt flag may not be really cleared before CPU exit ISR. In such case, user should add DSB instruction right after instruction to clear interrupt flag.

Table 4-2. CM7 domain interrupt summary

IRQ	Module	Logic	Description
0	eDMA	OR	eDMA Channel 0 Transfer Complete
			eDMA Channel 16 Transfer Complete
1		OR	eDMA Channel 1 Transfer Complete
			eDMA Channel 17 Transfer Complete
2		OR	eDMA Channel 2 Transfer Complete
			eDMA Channel 18 Transfer Complete
3		OR	eDMA Channel 3 Transfer Complete
			eDMA Channel 19 Transfer Complete
4		OR	eDMA Channel 4 Transfer Complete
			eDMA Channel 20 Transfer Complete
5		OR	eDMA Channel 5 Transfer Complete
			eDMA Channel 21 Transfer Complete
6		OR	eDMA Channel 6 Transfer Complete
			eDMA Channel 22 Transfer Complete
7		OR	eDMA Channel 7 Transfer Complete
			eDMA Channel 23 Transfer Complete
8		OR	eDMA Channel 8 Transfer Complete
			eDMA Channel 24 Transfer Complete
9		OR	eDMA Channel 9 Transfer Complete
			eDMA Channel 25 Transfer Complete
10		OR	eDMA Channel 10 Transfer Complete
			eDMA Channel 26 Transfer Complete
11		OR	eDMA Channel 11 Transfer Complete
			eDMA Channel 27 Transfer Complete
12		OR	eDMA Channel 12 Transfer Complete
			eDMA Channel 28 Transfer Complete
13		OR	eDMA Channel 13 Transfer Complete
			eDMA Channel 29 Transfer Complete
14		OR	eDMA Channel 14 Transfer Complete
			eDMA Channel 30 Transfer Complete
15		OR	eDMA Channel 15 Transfer Complete
			eDMA Channel 31 Transfer Complete
16		-	Error Interrupt, Channels 0-15 / 16-31

Table continues on the next page...

Table 4-2. CM7 domain interrupt summary (continued)

IRQ	Module	Logic	Description
17	CM7	-	CTI trigger outputs
18		-	CTI trigger outputs
19		-	Core Platform exception IRQ
20	LPUART1 Serial6	OR	UART1 TX interrupt
			UART1 RX interrupt
21	LPUART2 Serial3	OR	UART2 TX interrupt
			UART2 RX interrupt
22	LPUART3 Serial2 (Serial4 on MM)	OR	UART3 TX interrupt
			UART3 RX interrupt
23	LPUART4 Serial4 (Serial2 on MM)	OR	UART4 TX interrupt
			UART4 RX interrupt
24	LPUART5 Serial8	OR	UART5 TX interrupt
			UART5 RX interrupt
25	LPUART6 Serial1	OR	UART6 TX interrupt
			UART6 RX interrupt
26	LPUART7 Serial7	OR	UART7 TX interrupt
			UART7 RX interrupt
27	LPUART8 Serial5	OR	UART8 TX interrupt
		OR	UART8 RX interrupt
28	LPI2C1 Wire	OR	LPI2C1 Interrupt master async
			LPI2C1 Interrupt slave async
			LPI2C1 Interrupt master
			LPI2C1 Interrupt slave
29	LPI2C2 Wire3 on MM	OR	LPI2C2 Interrupt master async
			LPI2C2 Interrupt slave async
			LPI2C2 Interrupt master
			LPI2C2 Interrupt slave
30	LPI2C3 Wire1 (Wire2 on MM)	OR	LPI2C3 Interrupt master async
			LPI2C3 Interrupt slave async
			LPI2C3 Interrupt master
			LPI2C3 Interrupt slave
31	LPI2C4 Wire2 (Wire1 on MM)	OR	LPI2C4 Interrupt master async
			LPI2C4 Interrupt slave async
			LPI2C4 Interrupt master
			LPI2C4 Interrupt slave
32	LPSP1 SPI2	-	LPSP1 interrupt request line to the core
33	LPSP12	-	LPSP1 interrupt request line to the core
34	LPSP13 SPI1	-	LPSP1 interrupt request line to the core
35	LPSP14 SPI	-	LPSP1 interrupt request line to the core

Table continues on the next page...

Table 4-2. CM7 domain interrupt summary (continued)

IRQ	Module	Logic	Description
36	FLEXCAN1	-	Combined interrupt of ini_int_busoff, ini_int_error, ipi_int_mbor, ipi_int_rxwarning, ipi_int_txwarning and ipi_int_wakein, etc.
37	FLEXCAN2	-	Combined interrupt of ini_int_busoff, ini_int_error, ipi_int_mbor, ipi_int_rxwarning, ipi_int_txwarning and ipi_int_wakein, etc.
38	CM7	-	FlexRAM address out of range Or access hit IRQ
39	KPP	-	Keypad Interrupt
40	TSC_DIG	-	TSC interrupt
41	GPR_IRQ	-	Used to notify cores on exception condition while boot
42	LCDIF	-	LCDIF Sync Interrupt
43	CSI	-	CSI interrupt
44	PXP	-	PXP interrupt
45	WDOG2	-	Watchdog Timer reset
46	SNVS_HP_WRAPPER	-	SNVS Functional Interrupt
47		-	SNVS Security Interrupt
48	SNVS_LP_WRAPPER	OR	ON-OFF button press shorter than 5 secs (pulse event)
	SNVS_HP_WRAPPER		ON-OFF button press shorter than 5 secs (pulse event)
49	CSU	-	CSU Interrupt Request 1. Indicates to the processor that one or more alarm inputs were asserted
50	DCP	-	Combined DCP channel interrupts (except channel 0) and CRC interrupt
51		-	IRQ of DCP channel 0
52		-	Reserved
53	TRNG	-	TRNG interrupt
54	Reserved	-	Reserved
55	BEE	-	BEE IRQ
56	SAI1	OR	SAI RX interrupt
			SAI RX async interrupt
			SAI TX interrupt
			SAI TX async interrupt
57	SAI2	OR	SAI RX interrupt
			SAI RX async interrupt
			SAI TX interrupt
			SAI TX async interrupt
58	SAI3	-	SAI RX interrupt
59		-	SAI TX interrupt
60	SPDIF	OR	SPDIF Rx interrupt
			SPDIF Tx interrupt
61	PMU	-	Brown-out event on either the 1.1, 2.5 or 3.0 regulators.
62	Reserved	-	Reserved

Table continues on the next page...